AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
MECHATRONICS ENGINEERING DEPARTMENT

Fall 2026

MCT 331s: Mechatronic System Integration

Smart Home Demo Project

Overview:

In this project, teams composed of 3 to 4 students will design and build a smart home prototype with basic features that demonstrate the integration of mechanical, electrical, and computer systems. The smart home prototype should use Arduino board and include the following functions:

1. **Main Door Access**
   The main door opens when the correct code is entered via a keypad and closes automatically after a preset delay.

2. **Security System**
   When the security mode is activated:
   - The system triggers an alarm if the window is opened.
   - The exterior light turns on automatically when motion is detected near the main door.

3. **Automatic Indoor Lighting System**
   The indoor light turns on automatically when the ambient light is below a set threshold or when the user presses a manual override button.

4. **Climate Control System**
   Activates a fan when the temperature exceeds a user-defined threshold set by an adjustable knob.

5. **LCD Feedback Display**
   A single LCD module provides real-time system feedback, e.g.:
   - Door status
   - Security mode
   - Temperature readings

Learning Goals:

1. Applying all the acquired knowledge regarding basic sensors.
2. Fabrication of a compact functioning electrical system (PCBs, wiring, etc.).
3. Implement sensor-actuator integration with a microcontroller.
4. Develop simple control algorithms (decision-making and automation).
5. Demonstrate practical prototyping skills in wiring, coding, and mechanical assembly.

Online Submissions:

Each team will be obliged to upload a technical report online prior to the final project submission day.The report must be comprehensive and structured to document the integration of mechanical, electrical, and computer systems. Sections include:

- **Bill of Material**
  a. All components classification (Sensor, Actuator, Information processing and mechanical System).
  b. Function.
  c. Material of mechanical components.

- **Drawings**
  CAD Construction drawings showing the physical location of the main sensors and actuators.

- **Electrical and Electronic System**
  a. PCB layouts
  b. System interconnection diagram
  c. Circuit simulation results (if any)

- **Control System**
  a- Flow charts for all control algorithms.
  b- Code Documentation.

Demonstration Video

Teams will also be obliged to submit a short video (2-3 minutes) demonstrating the teamwork during the project phases short clips and the most important integration phases.

Project Progress: **21 & 22 Oct.**

Project Submission: **Week 9 (14-20 Nov.)**